EXP 11: APPLICATION DEPLOYMENT AND MIGRATION IN ZOHO CREATOR

AIM

To study and perform the deployment of the Library Book Issue Management application from the development stage to the live (production) stage, and to understand how the application structure and data are migrated (exported and imported) between environments and accounts in Zoho Creator.

PROCEDURE

1. **Understand the environments**
   - Zoho Creator separates an application into environments so that changes can be tested before users see them.
   - **Development** is where the application is built and edited; **Stage** is used for testing the changes; **Production** is the live version used by the librarian and students.
   - The availability of the Stage and Production environments depends on the Zoho Creator plan.

Environment	Purpose	Who uses it
Development	Build and edit forms, reports, pages and workflows	Developer
Stage	Test the changes with sample data before release	Developer and testers
Production	Live application with real records	Librarian and students

2. **Complete the pre-deployment checklist**
   - Finish all forms, reports, dashboards, pages, workflows, functions and blueprints.
   - Test every workflow (form, scheduled, batch) with sample data and fix the Deluge errors.
   - Check the roles and permissions for Librarian and Student, and remove unused test records and test users.
   - Take a backup by exporting the application and its data before making the changes live.
3. **Deploy the application**
   - Open the application in the edit mode of the Development environment.
   - Use the **Deploy** option to review the list of changes (new and modified components) and select the target environment.
   - Deploy the changes to Stage, test them, and then deploy to Production; users in Production see only the deployed version.
4. **Export the application**
   - From the application options, choose **Export** to download the application as a **.ds** file.
   - The .ds file contains the structure of the application: forms, fields, reports, pages, workflows, functions, roles and permissions. The records (data) are not included.
5. **Import (migrate) the application into another account**
   - In the target account, create a new application using the **Import** option and upload the .ds file.
   - A copy of the application is created with all its forms, reports, pages and workflows, which can be edited and deployed independently.
6. **Migrate the data**
   - Export the records from the reports in CSV or Excel format.
   - In the new application, use the **Import Data** option, upload the file and map each column to the matching form field.
   - Import the parent forms first (for example, Book_Catalog) and then the dependent forms (Book Issue Request) so that lookup values exist.

- Check the date and time formats (for example, dd-MMM-yyyy) and compare the record count before and after the import.
- Data from other systems can also be migrated by importing Excel or CSV files, or by using Deluge and API integration.

7. **Complete the post-deployment configuration**
   - Re-create the connections and update the application variables used in Deluge scripts.
   - Add the users again, assign the Librarian and Student roles and permissions, and share the application.
   - Enable the scheduled and batch workflows and check the email notification settings and the sender address.
8. **Verify the deployment**
   - Open the Production application, submit a test Book Issue Request and check the email notification and the log details.
   - Check the reports, dashboard and pages, and confirm that students can only access the permitted forms.
9. **Maintain versions and rollback**
   - Keep dated .ds backups of each released version of the application.
   - If a problem occurs, restore the previous version by importing the earlier .ds file or by redeploying the earlier version.
   - Make all later changes in Development first, test them in Stage, and then deploy to Production.

Deployment and Migration - Comparison

Point	Deployment	Migration
Meaning	Moving changes from one environment to another in the same application	Moving the application or its data to another account or system
Method	Deploy option (Development to Stage to Production)	Export and import of the .ds file and the data files
What moves	Changes in forms, reports, pages and workflows	Application structure (.ds) and records (CSV or Excel)
Main care	Test in Stage before Production	Import parent forms first and check the record count

Common Problems During Migration and Their Solutions

Problem	Cause	Solution
Lookup values missing after import	Dependent form imported before the parent form	Import the parent form (Book_Catalog) first, then the dependent form
Records rejected during import	Wrong date format or a mandatory field left empty	Correct the format (dd-MMM-yyyy) and fill the required fields in the file
Workflows or emails not working	Connections, variables or schedules not set again	Re-create the connections, update the variables and enable the workflows
Users cannot see the forms	Roles and permissions not assigned in the new application	Assign the Librarian and Student roles and permissions to the users

RESULT

The deployment and migration process in Zoho Creator was studied successfully. The Library Book Issue Management application can be tested and released through the Development, Stage and Production environments, and its structure and data can be exported, imported and verified in another account without loss of data. Following the pre-deployment checklist, post-deployment configuration and backups ensures a safe and reliable release.